

CHAPTER 2

Measurement

Roadmap



1. Measuring GDP

- (1) National Income Accounts

- (2) Nominal and Real GDP

- (3) Price Indices

2. Savings, Wealth, and Capital

3. Labor Market Measurement

1. Measuring GDP



- National Income Account Calculation
(Three approaches – Product, Expenditure, Income)
- Example with Inventory Investment
- Example with International Trade
- GNP
- Components of Aggregate Expenditure
- Issues in Measuring GDP

Table 2.9 Gross Domestic Product for 2008

Component of GDP	\$Billions	% of GDP
GDP	14,264.6	100.0
Consumption	10,057.9	70.5
Durables	1023.2	7.2
Nondurables	2965.1	20.8
Services	6069.6	42.6
Investment	1993.5	14.0
Fixed Investment	2040.5	14.3
Nonresidential	1552.8	10.9
Residential	487.7	3.4
Inventory Investment	−47.0	−0.3
Net Exports	−669.2	−4.7
Exports	1859.4	13.0
Imports	2528.6	17.7
Government Expenditures	2882.4	20.2
Federal Defense	734.9	5.2
Federal Nondefense	337.0	2.4
State and Local	1810.4	12.7

Table 2.9 Gross Domestic Product for 2011

Component of GDP	\$Billions	% of GDP
GDP	15,094.0	100.0
Consumption	10,726.0	71.1
Durables	1,162.9	7.7
Nondurables	2,483.7	16.4
Services	7,079.4	46.9
Investment	1,916.2	12.7
Fixed Investment	1,870.0	12.4
Nonresidential	1,532.5	10.2
Residential	337.5	2.2
Inventory Investment	46.3	0.3
Net Exports	-578.7	-3.8
Exports	2,085.5	13.8
Imports	2,664.2	17.7
Government Expenditures	3,030.6	20.1
Federal Defense	824.9	5.5
Federal Nondefense	407.9	2.7
State and Local	1,797.7	11.9

Table 2.9 Gross Domestic Product for 2015

Component of GDP	\$Billions	% of GDP
GDP	17,937.8	100.0
Consumption	12,267.9	68.4
Durables	1,328.8	7.4
Nondurables	2,649.8	14.8
Services	8,289.3	46.2
Investment	3,017.8	16.8
Fixed Investment	2,911.3	16.2
Nonresidential	2,302.4	12.8
Residential	608.9	3.4
Inventory Investment	106.5	0.6
Net Exports	531.9	3.0
Exports	2,253.0	12.6
Imports	2,784.9	15.5
Government Expenditures	3,184.0	17.8
Federal Defense	740.9	4.1
Federal Nondefense	483.9	2.7
State and Local	1,959.3	10.9

Table: Gross Domestic Product for 2021

Component of GDP	\$Billions	% of GDP
Gross domestic product	23,315.1	100.0
Consumption expenditures	15,902.6	68.2
Durable goods	2,060.2	8.8
Nondurable goods	3,436.3	14.7
Services	10,406.1	44.6
Gross private domestic investment	4,113.5	17.6
Fixed investment	4,132.6	17.7
Nonresidential	3,025.0	13.0
Residential	1,107.6	4.8
Change in private inventories	- 19.1	-0.1
Net exports of goods and services	- 861.7	-3.7
Exports	2,539.6	10.9
Imports	3,401.4	14.6
Government expenditures	4,160.7	17.8
Federal	1,609.2	6.9
National defense	904.0	3.9
Nondefense	705.1	3.0
State and local	2,551.6	10.9

GDP for 2008 in Korea

source: ECOS

ACCOUNT (10.3.1.1 국내총생산과 지출)	₩Billions	% of GDP
Expenditure on Gross Domestic Product (국내총생산에 대한 지출)	1,104,492.2	100.0%
Private Final Consumption Expenditure (민간최종소비 지출)	579,053.4	52.4%
Government Final Consumption Expenditure (정부최종소비 지출)	161,751.2	14.6%
Gross Fixed Capital Formation (총고정자본 형성)	346,612.1	31.4%
Changes in Inventories and Acquisitions less Disposals of Valuables (재고증감 및 귀중품 순취득)	18,074.7	1.6%
Net Exports (순수출)	-118.6	-0.0%
Exports of Goods and Services (재화와서비스의 수출)	551,819.9	50.0%
(less)Imports of Goods and Services ((공제)재화와 서비스의 수입)	551,938.5	50.0%
Statistical Discrepancy (통계상불일치)	-880.5	0.0%

GDP for 2011 in Korea

source: ECOS

ACCOUNT (10.3.1.1 국내총생산과 지출)	₩Billions	% of GDP
Expenditure on Gross Domestic Product (국내총생산에 대한 지출)	1,332,681.0	100.0%
Private Final Consumption Expenditure (민간최종소비 지출)	679,141.5	51.0%
Government Final Consumption Expenditure (정부최종소비 지출)	194,381.2	14.6%
Gross Fixed Capital Formation (총고정자본 형성)	403,045.3	30.2%
Changes in Inventories and Acquisitions less Disposals of Valuables (재고증감 및 귀중품 순취득)	36,190.8	2.7%
Net Exports (순수출)	19,922.2	1.5%
Exports of Goods and Services (재화와서비스의 수출)	742,936.0	55.7%
(less)Imports of Goods and Services ((공제)재화와 서비스의 수입)	723,013.8	54.3%
Statistical Discrepancy (통계상불일치)	0.0	0.0%

GDP for 2015 in Korea

source: ECOS

ACCOUNT (10.3.1.1 국내총생산과 지출)	₩Billions	% of GDP
Expenditure on Gross Domestic Product (국내총생산에 대한 지출)	1,564,123.0	100.0%
Private Final Consumption Expenditure (민간최종소비 지출)	771,239.2	49.3%
Government Final Consumption Expenditure (정부최종소비 지출)	234,766.4	15.0%
Gross Fixed Capital Formation (총고정자본 형성)	458,419.9	29.3%
Changes in Inventories and Acquisitions less Disposals of Valuables (재고증감 및 귀중품 순취득)	-6,104.8	-0.4%
Net Exports (순수출)	108,882.7	7.0%
Exports of Goods and Services (재화와서비스의 수출)	709,122.0	45.3%
(less)Imports of Goods and Services ((공제)재화와 서비스의 수입)	600,239.3	38.4%
Statistical Discrepancy (통계상불일치)	-3,079.4	-0.2%

GDP for 2021 in Korea

source: ECOS

ACCOUNT (10.3.1.1 국내총생산과 지출)	₩Billions	% of GDP
Expenditure on Gross Domestic Product (국내총생산에 대한 지출)	2,071,658.0	100.0%
Private Final Consumption Expenditure (민간최종소비 지출)	955,779.6	46.1%
Government Final Consumption Expenditure (정부최종소비 지출)	376,218.8	18.2%
Gross Fixed Capital Formation (총고정자본 형성)	654,034.3	31.6%
Changes in Inventories and Acquisitions less Disposals of Valuables (재고증감 및 귀중품 순취득)	10,760.3	0.5%
Net Exports (순수출)	74,137.4	3.6%
Exports of Goods and Services (재화와서비스의 수출)	870,831.7	42.0%
(less)Imports of Goods and Services ((공제)재화와 서비스의 수입)	796,694.3	38.5%
Statistical Discrepancy (통계상불일치)	727.5	0.04%

GDP(Gross Domestic Product)



- GDP: The dollar value of final output produced during a given period of time within the borders of a country.
- Data: BEA(Bureau of Economic Analysis) for the U.S.
Statistics Korea or ECOS for Korea
- GNP(Gross National Product): the dollar value of final output produced during a given period of time by domestic factors regardless of whether the production takes place within a country or not.

- BEA (<https://www.bea.gov/>) >> Tools >> Interactive Data (<https://www.bea.gov/itable/>) >> Gross Domestic Product/ Personal Income >> Interactive Data Tables >> Section 1, Table 1.1.5 >> Modify (change to annual table)
- ECOS (<https://ecos.bok.or.kr/#/>) >> National Accounts >> 2.1. National Income and Product >> 2.1.3. Consolidated Accounts for the Nation >> 2.1.3.1. Gross Domestic Product and Expenditure >> 2.1.3.1.1. Gross Domestic Product and Expenditure >> Add items selected >> View the list

- Three approaches to measuring GDP
 - Product Approach
 - Expenditure Approach
 - Income Approach
- Income-Expenditure Identity with $Y = zF(K, N)$:

$$Y = C + I + G + NX$$

where Y is total GDP, total output, and total income.

Example of Measuring GDP

An Economy with

- A coconut producer
 - Each coconut costs \$2 dollars.
 - 6 million coconuts are sold to the restaurant.
 - 4 million coconuts are sold to consumers.
- A restaurant
- Consumers
- A government

Table 2.1 Coconut Producer

Total Revenue	\$20 million
Wages	\$5 million
Interest on Loan	\$0.5 million
Taxes	\$1.5 million

Table 2.2 Restaurant

Total Revenue	\$30 million
Cost of Coconuts	\$12 million
Wages	\$4 million
Taxes	\$3 million

- Coconut producer's after-tax profit
= \$20mil - \$5mil - \$0.5mil - \$1.5mil = \$13mil
- A restaurant's after-tax profit
= \$30mil - \$12mil - \$4mil - \$3mil = \$11mil

Table 2.5 Consumers

Wage Income	\$14.5 million
Interest Income	\$0.5 million
Taxes	\$1 million
Profits Distributed to Producers	\$24 million

Table 2.4 Government

Tax Revenue	\$5.5 million
Wages	\$5.5 million

For consumers,

- Wage Income = \$5mil + \$4mil + \$5.5mil = \$14.5mil
- Profits = \$13mil + \$11mil = \$24mil

The Product Approach

- GDP is calculated as the sum of value added to goods and services across all productive units in the economy.

“Add the value of all final goods and services produced in the economy – the value of all intermediate goods used in production to obtain total value added.”

- Value added – coconuts from producers = \$20mil
 - Value added – restaurant food = \$30mil - \$12mil = \$18mil
 - Value added – government = \$5.5mil
-

- $\text{GDP} = \$20\text{mil} + \$18\text{mil} + \$5.5\text{mil} = \43.5mil

The Expenditure Approach

- GDP is calculated as total spending on final goods and services production in the economy.
- Total Expenditure = $C + I + G + NX$
 - C is consumption
 - I is investment
 - G is government spending
 - NX is net exports that equals total exports minus total imports

- In this example, $I = NX = 0$.
 - Consumers spend \$8mil on coconuts and \$30mil at the restaurant. Thus, $C = \$38\text{mil}$
 - $G = \$5.5\text{mil}$
-

- $GDP = \$38\text{mil} + \$5.5\text{mil} = \$43.5\text{mil}$

The Income Approach

- GDP is calculated as adding up all income received by economic agents contributing to production, where income includes the profit made by firms such as
 - (1) Compensation of employees
(for example: wage, salaries, and benefits)
 - (2) Proprietor's income(self-employed firm owners)
 - (3) Rental income
 - (4) Corporate profits
 - (5) Net interest
 - (6) Indirect business/producer's taxes
 - (7) Depreciation

- Wage income = \$5mil + \$4mil + \$5.5mil = \$14.5mil
 - After-tax profits = \$13mil + \$11mil = \$24mil
 - Interest income = \$0.5mil
 - Taxes (Only producers) = \$1.5mil + \$3mil = \$4.5mil
-

- GDP = \$14.5mil + \$24mil + \$0.5mil + \$4.5mil = \$43.5mil

Example with Inventory Investment

- Stocks of inventories
 - (1) Finished goods
 - (2) Goods in process
 - (3) Raw materials
- Example: Suppose the coconut produced 13 mil coconuts. Then, 3 mil are not sold and stored as inventory. Everything else is same as before. Coconut producer's after-tax profit is \$19mil ($= \$26\text{mil} - \$5\text{mil} - \$0.5\text{mil} - \1.5mil).

The Product Approach:

- Value added – coconuts from producers = \$26mil
 - Value added – restaurant food = \$30mil - \$12mil = \$18mil
 - Value added – government = \$5.5mil
-

- $GDP = \$26\text{mil} + \$18\text{mil} + \$5.5\text{mil} = \49.5mil

The Expenditure Approach:

- In this example, $NX = 0$.
 - $I = \$2 \times 3\text{mil} = \6mil .
 - Consumers spend \$8mil on coconuts and \$30mil at the restaurant. Thus, $C = \$38\text{mil}$
 - $G = \$5.5\text{mil}$
-

- $GDP = \$38\text{mil} + \$6\text{mil} + \$5.5\text{mil} = \49.5mil

The Income Approach:

- Wage income = \$5mil + \$4mil + \$5.5mil = \$14.5mil
- After-tax profits = \$19mil + \$11mil = \$30mil
- Interest income = \$0.5mil
- Taxes (Only producers) = \$1.5mil + \$3mil = \$4.5mil

○ $GDP = \$14.5\text{mil} + \$30\text{mil} + \$0.5\text{mil} + \$4.5\text{mil} = \$49.5\text{mil}$

Example with International Trade

- Allow international trade and no inventories
- Example: Suppose the restaurant imports 2 mil coconuts from other islands at \$2 each, in addition to the coconuts purchased from the domestic coconut producer and that all of these coconuts are used in the restaurant. Everything else is same as before without inventories. The restaurant's after-tax profit is \$7mil ($= \$30\text{mil} - \$12\text{mil} - \$4\text{mil} - \$4\text{mil} - \3mil).

The Product Approach:

- Value added – coconuts from producers = \$20mil
 - Value added – restaurant food = \$30mil - \$16mil = \$14mil
 - Value added – government = \$5.5mil
-

- $GDP = \$20\text{mil} + \$18\text{mil} + \$5.5\text{mil} = \39.5mil

The Expenditure Approach:

- In this example, $NX = -\$4\text{mil.}$
 - $I = 0.$
 - Consumers spend \$8mil on coconuts and \$30mil at the restaurant. Thus, $C = \$38\text{mil}$
 - $G = \$5.5\text{mil}$
-

- $GDP = \$38\text{mil} + \$5.5\text{mil} - \$4\text{ mil} = \39.5mil

The Income Approach:

- Wage income = \$5mil + \$4mil + \$5.5mil = \$14.5mil
- After-tax profits = \$13mil + \$7mil = \$20mil
- Interest income = \$0.5mil
- Taxes (Only producers) = \$1.5mil + \$3mil = \$4.5mil

○ $GDP = \$14.5\text{mil} + \$20\text{mil} + \$0.5\text{mil} + \$4.5\text{mil} = \$39.5\text{mil}$

GNP(Gross National Product)

- GNP(Gross National Product): the dollar value of final output produced during a given period of time by domestic factors regardless of whether the production takes place within a country or not.
- Relationship with GDP: $GNP = GDP + NFP$

- NFP is net factor payments from abroad.
 - (1) Factor payments(FP): The income people receive for supplying the factors of production including rent, wage, interest, and profit.
 - (2) $NFP = FP \text{ from abroad} - FP \text{ to abroad}$
 - (3) Example:
 - 1) Wages of Korean workers in Hyundai plant in Georgia, U.S. (FP from abroad)
 - 2) Wages of US workers in Nike plant in Korea (FP to abroad)

Components of Aggregate Expenditure

Table 2.9 Gross Domestic Product for 2008

Component of GDP	for 2008		for 2011		for 2015	
	\$Billions	% of GDP	\$Billions	% of GDP	\$Billions	% of GDP
GDP	14,264.6	100.0	15,094.0	100.0	17,937.8	100.0
Consumption	10,057.9	70.5	10,726.0	71.1	12,267.9	68.4
Durables	1,023.2	7.2	1,162.9	7.7	1,328.8	7.4
Nondurables	2,965.1	20.8	2,483.7	16.4	2,649.8	14.8
Services	6,069.6	42.6	7,079.4	46.9	8,289.3	46.2
Investment	1,993.5	14.0	1,916.2	12.7	3,017.8	16.8
Fixed Investment	2,040.5	14.3	1,870.0	12.4	2,911.3	16.2
Nonresidential	1,552.8	10.9	1,532.5	10.2	2,302.4	12.8
Residential	487.7	3.4	337.5	2.2	608.9	3.4
Inventory Investment	-47.0	-0.3	46.3	0.3	106.5	0.6
Net Exports	-669.2	-4.7	-578.7	-3.8	531.9	3.0
Exports	1,859.4	13.0	2,085.5	13.8	2,253.0	12.6
Imports	2,528.6	17.7	2,664.2	17.7	2,784.9	15.5
Government Expenditures	2,882.4	20.2	3,030.6	20.1	3,184.0	17.8
Federal Defense	734.9	5.2	824.9	5.5	740.9	4.1
Federal Nondefense	337.0	2.4	407.9	2.7	483.9	2.7
State and Local	1,810.4	12.7	1,797.7	11.9	1,959.3	10.9

- Consumption: around 70% of GDP.
 - Durable Goods: lasted for a while
example: new automobiles, appliances, and furniture
 - Nondurable Goods: consumed within the period
example: food, clothing, stationary
 - Services: nontangible items
example: haircuts, hotels, and education

Note that used items are not included in GDP, but the service provided (by a dealer) would be included.

- Investment: more than 12% of GDP.
 - Goods produced in the current period but not consumed in the current period
 - Types
 - (i) Fixed Investment: production of capital such as plant, equipment, and housing.
 - a) Nonresidential Investment
 - b) Residential Investment
 - (ii) Inventory Investment: goods that are essentially put into storage.

○ Government Expenditures

- Expenditures by federal, state, and local government on final goods and services
- Components: federal defense spending, federal non-defense spending, state & local spending

Note that government expenditure does not include 'transfers' (example: social security payments and unemployment insurance payments). They are simply money transfers from one to another.

Problems of Measuring GDP

- Inaccurate Measure of Welfare: Not account for the distribution of income across economic individuals.
- Leaves out all nonmarket activity including home production
- Leaves out underground economy
- Others including quality improvement, change in leisure time, measure of government expenditure

2. Nominal & Real GDP and Price Indices

- Economists would like to compare GDP data in different time periods for many purposes such as economic growth, standard of living, and inflation.
- Also, they are interested in the adjustment of the effect of inflation on GDP.

To do list:

- Nominal and Real GDP
- Measures of the Price Level
- Problems with measuring real GDP and the price level

- A Price Index: A weighted average of the prices of a set of the goods and services produced in the economy over a period of time.

If the price index includes prices of all goods and services, then that price index is a measure of the general price level.

- The Inflation Rate: The rate of change in the price level from one period of time to another.

Example of Nominal & Real GDP

Table 2.10 Data for Real GDP Example

	Apples	Oranges
Quantity in Year 1	$Q_1^a = 50$	$Q_1^o = 100$
Price in Year 1	$P_1^a = \$1.00$	$P_1^o = \$0.80$
Quantity in Year 2	$Q_2^a = 80$	$Q_2^o = 120$
Price in Year 2	$P_2^a = \$1.25$	$P_2^o = \$1.60$

(1) Nominal GDP(NGDP)

$$\begin{aligned} 1) \text{ Year 1: } NGDP_1 &= P_1^a Q_1^a + P_1^o Q_1^o \\ &= 50 * \$1.00 + 100 * \$0.80 = \$130 \end{aligned}$$

$$\begin{aligned} 2) \text{ Year 2: } NGDP_2 &= P_2^a Q_2^a + P_2^o Q_2^o \\ &= 80 * \$1.25 + 120 * \$1.60 = \$292 \end{aligned}$$

3) The percentage change in NGDP from Year1 to Year2

$$\left(\frac{NGDP_2}{NGDP_1} - 1 \right) * 100 = \left(\frac{292}{130} - 1 \right) * 100 = 125\%$$

Question: How much of this increase in NGDP is accounted for by inflation and how much by an increase in the real quantity of aggregate output produced?

(2) Real GDP(RGDP)

1) Suppose year 1 is the base year. Then,

$$a) RGDP_1 = NGDP_1 = P_1^a Q_1^a + P_1^o Q_1^o = \$130$$

$$b) RGDP_2 = P_1^a Q_2^a + P_1^o Q_2^o \\ = 80 * \$1.00 + 120 * \$0.80 = \$176$$

c) The percentage change in RGDP from Year1 to Year2

$$\left(\frac{RGDP_2}{RGDP_1} - 1 \right) * 100 = \left(\frac{176}{130} - 1 \right) * 100 = 35.4\%$$

2) Suppose year 2 is the base year. Then,

$$\begin{aligned} \text{a) } RGDP_1 &= P_2^a Q_1^a + P_2^o Q_1^o \\ &= 50 * \$1.25 + 100 * \$1.60 = \$222.50 \end{aligned}$$

$$\text{b) } RGDP_2 = NGDP_2 = P_2^a Q_2^a + P_2^o Q_2^o = \$292$$

c) The percentage change in RGDP from Year1 to Year2

$$\left(\frac{RGDP_2}{RGDP_1} - 1 \right) * 100 = \left(\frac{292}{222.50} - 1 \right) * 100 = 31.2\%$$

(3) Chain-Weighting Approach

The problem of changing relative prices would be severe if calculating RGDP in a year far from a base year. Thus, the Fisher index is a useful tool to ease the matter.

The Fisher Index: The chain-weighted ratio of RGDP in the two years is a geometric average of the ratios calculated using each of years 1 and 2 as base years.

The Fisher Index is given by

$$g_c = \sqrt{g_1 * g_2} = \sqrt{1.354 * 1.312} = 1.333$$

where $g_i = \frac{RGDP_2}{RGDP_1}$ and i is the base year. Thus, the percentage change in RGDP from Year1 to Year2 is 33.3%.

For n years, $g_c = \sqrt[n]{g_1 * g_2 * \dots * g_n}$ or $\ln(g_c) = \frac{1}{n} \sum_{i=1}^n \ln(g_i)$.

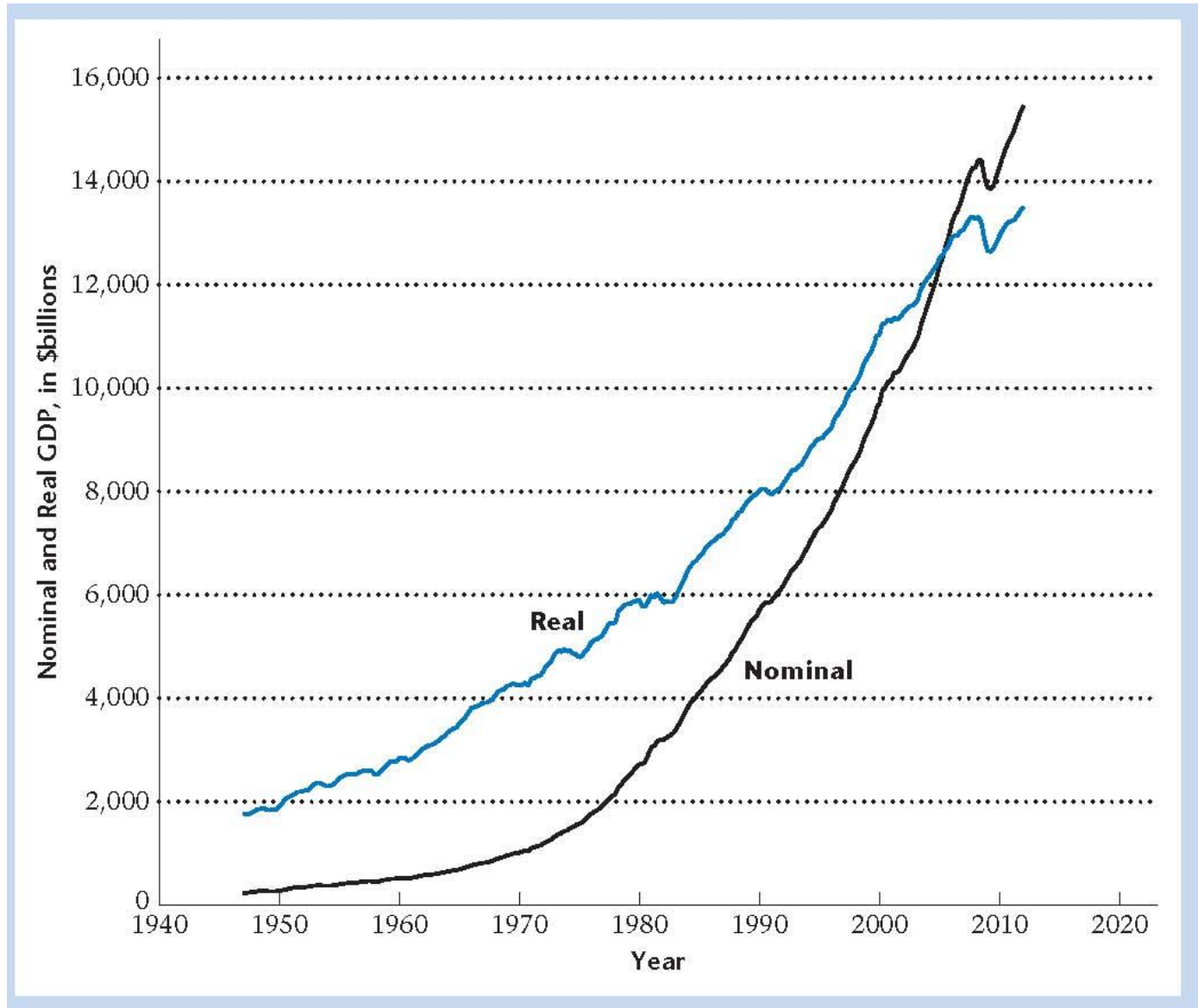
[Application of the Fisher index]

- Given g_c , we can calculate RGDP in terms of the dollars of any year.

Example: Suppose $g_c=1.333$ and $RGDP_1=\$130$. Then, we can find out $RGDP_2 = \$130 \times 1.333 = \173.29 .

- In Figure 2.1, RGDP grows at a lower rate than NGDP due to positive inflation.

Figure 2.1 NGDP and Chain-Weighted RGDP (Base year: 2005)



The Price Level and Example



- Two measures of the price level
 - Implicit GDP price deflator: a price of aggregate output
 - Consumer price index (CPI): a cost of living

- Implicit GDP Price Deflator

$$= \frac{NGDP}{RGDP} * 100$$

$$= \frac{\text{Cost of current year } Q \text{ at current year } P}{\text{Cost of current year } Q \text{ at base year } P} * 100$$

Table 2.11 Implicit GDP Price Deflators, Example

	Year 1	Year 2	% Increase
Year 1 = base year	100	165.9	65.9
Year 2 = base year	58.4	100	71.2
Chain-weighting	100	168.5	68.5

- CPI at current year

$$= \frac{\text{Cost of base year } Q \text{ at current year } P}{\text{Cost of base year } Q \text{ at base year } P} * 100,$$

implicitly assuming consumers do not change their buying habits when relative price changes occur.

Example: Suppose year 1 is the base year. Then,

$$\text{a) } CPI_1 = \frac{\$130}{\$130} * 100 = 100$$

$$\text{b) } CPI_2 = \frac{\$222.5}{\$130} * 100 = 171.2$$

- c) The percentage increase in CPI from year 1 to year 2 is 71.2%(=171.2-100).

Figure 2.2

Inflation Rate from CPI and from Implicit GDP Price Deflator

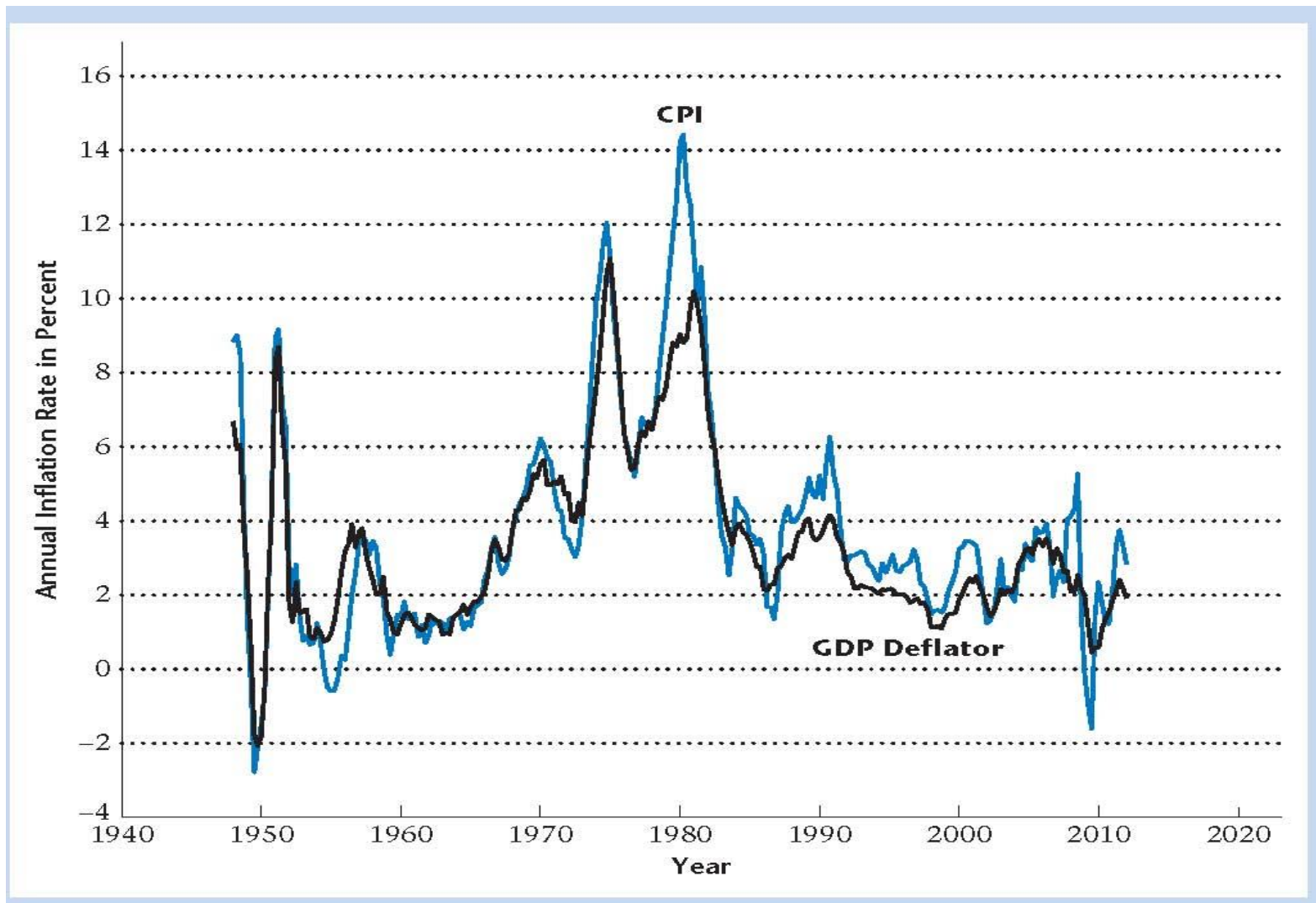
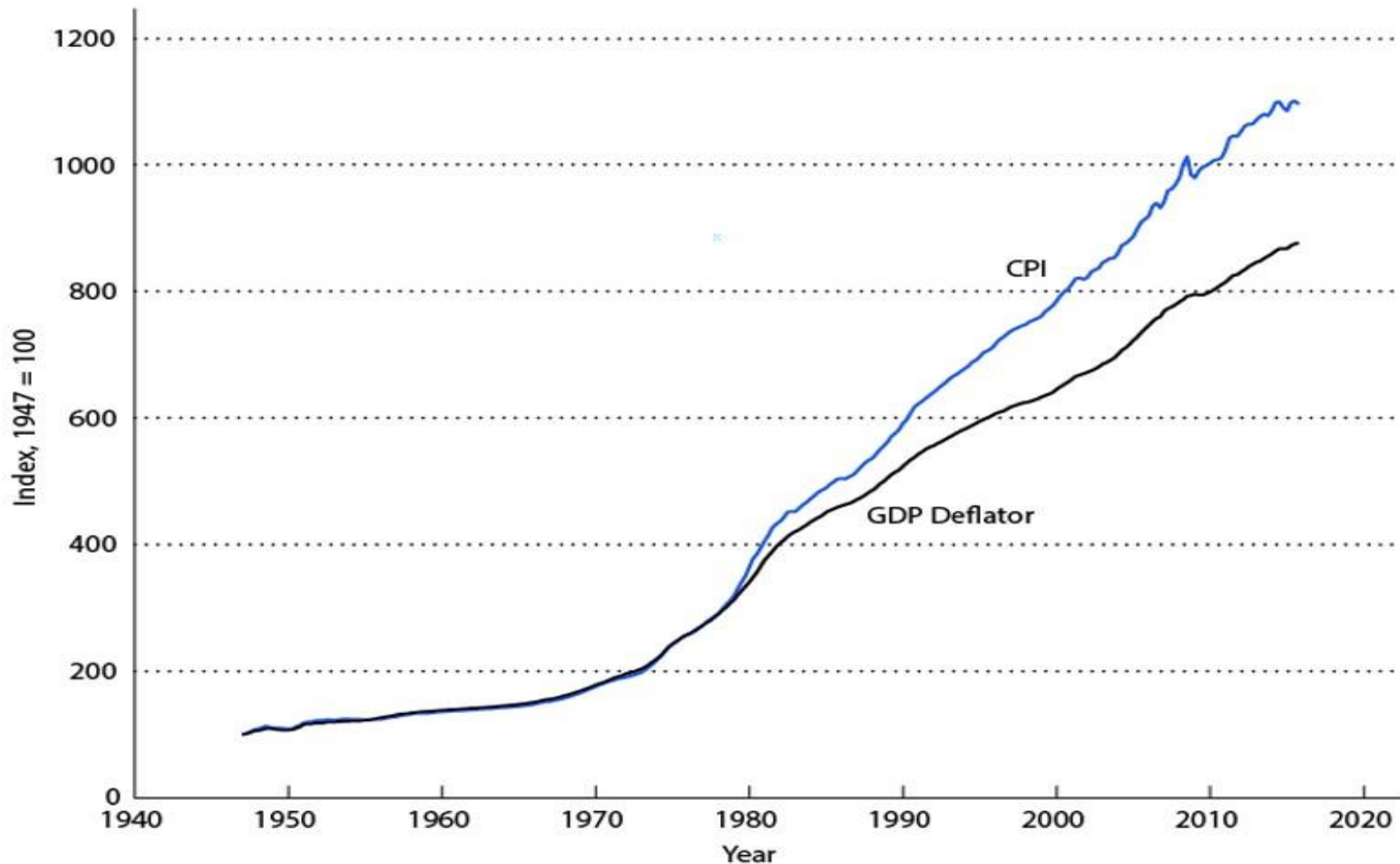


Figure 2.3

The Price Level as Measured by CPI and Implicit GDP Price Deflator



- Problems with measuring RGDP and the Price level
 - Changes in relative prices can also introduce severe bias in how the CPI measures inflation.
(See 'Critiquing the Consumer Price Index' by Kevin Kliesen for more discussion about it.)
 - Changes in the quality of goods overtime.
 - Introducing new goods.

3. Savings, Wealth, and Capital

- Flows and Stocks

- Flow: a rate per unit time
example] GDP, consumption, investment, government spending, net exports, savings
- Stock: the quantity in existence of some object at a point in time
example] houses, land, wealth, capital

○ Savings

(1) Private Savings

Given $Y^d = Y + NFP + TR + INT - T$,

$$S^p = Y^d - C = Y + NFP + TR + INT - T - C$$

where Y^d is disposable income, TR is transfers from gov't to the private sector, INT is interest on gov't debt, and T is taxes.

(2) Gov't Savings

$$S^g = T - TR - INT - G (= -D),$$

where D is gov't deficit.

(3) National Savings

$$\begin{aligned} S &= S^p + S^g = Y + \text{NFP} - C - G \\ &= C + I + G + \text{NX} + \text{NFP} - C - G \\ &= I + \text{NX} + \text{NFP} \\ &= I + \text{CA} \end{aligned}$$

where CA is current account surplus that is a measure of the balance of trade.

4. Labor Market Measurement

- Bureau of Labor Statics for US data
- Statistics Korea or ECOS for Korea data
- Employed: those who worked part-time or full-time during the past week
- Unemployed: those who were not employed during the past week but actively searched for work at some time during the last four weeks.
- Not in the labor force: those who are neither employed or unemployed.

- Labor Force = Employed + Unemployed
- Unemployment Rate
= # of Unemployed/Labor Force
- Participation Rate
= Labor Force/Total Working Age Population
- Employment/Population Ratio
= Total Employment/Total Working Age Population

- Labor Market Tightness

= Job Vacancy Rate/Unemployment Rate

(or = # of Job Vacancies/ # of Unemployed)

- Job Vacancy Rate

= # of Job Vacancies/[# of Job Vacancies + # of Employed]

- Unemployment rate is a measure of labor market tightness.
- Unemployment rate can be a mismeasure labor market tightness due to discouraged workers and no consideration for the intensity of job search